

Quiz 1 Solutions

CPSC 406 — Algorithm Analysis, Section 2

Chapman University · Spring 2026

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1. (2 points) (a) The correct answer is

$$L_3 = \{ w \in \Sigma^* \mid |w| \equiv 0 \pmod{3} \}.$$

The DFA starts in the accepting state p , and every symbol, whether a or b , moves one state forward around the 3-cycle. Therefore the state records the input length modulo 3.

The other choices are wrong because L_1 contains only words of length exactly 3, not all lengths divisible by 3. The choices L_2 and L_4 count only a 's, but this DFA moves on both a 's and b 's. “None of these” is wrong because L_3 is correct.

- (b) The correct answer is

All of these.

The word aab is accepted by the path

$$1 \xrightarrow{a} 1 \xrightarrow{a} 2 \xrightarrow{b} 3.$$

The word abc is accepted by the path

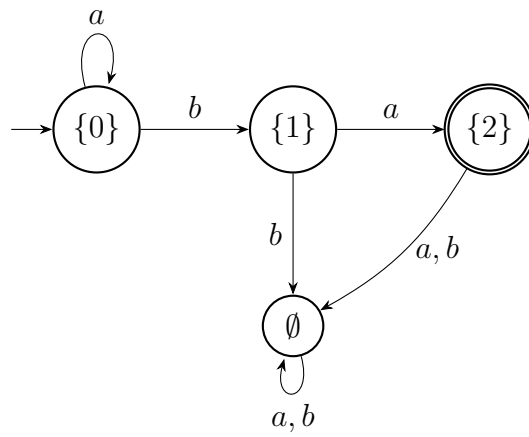
$$1 \xrightarrow{a} 1 \xrightarrow{b} 2 \xrightarrow{c} 2.$$

The word ccc is accepted by the path

$$1 \xrightarrow{c} 1 \xrightarrow{c} 1 \xrightarrow{c} 2.$$

States 2 and 3 are accepting, so all three listed words are accepted. Thus choosing only one word is too restrictive, and “None of these” is wrong.

2. (2 points) The subset construction gives the following DFA. The accepting DFA states are those subsets containing the accepting NFA state 2.



This DFA accepts exactly strings of the form a^*ba .